

A disproof of Conjecture 00000002348

An expected condition number is at least 1, never $m^{-1/2}$

1 The conjecture and its reading

Conjecture 00000002348 reads: *The expected condition number of the norm matrix of random interpolation on smooth varieties is $m^{-1/2}$ (m the sample count; a negative half power); the expectation comes from the logarithmic-potential integral of the determinant of the norm matrix.*

The claim is a value for $\mathbb{E}[\kappa(A)]$, where A is the (random) norm matrix built from m random samples. We refute two readings of “is $m^{-1/2}$ ”:

- (E) *exact*: $\mathbb{E}[\kappa(A_m)] = m^{-1/2}$ for the sample counts m ;
- (O) *order of magnitude*: $\mathbb{E}[\kappa(A_m)] \leq C m^{-1/2}$ for some constant C and all large m . This is implied by $\mathbb{E}[\kappa(A_m)] = C m^{-1/2}$, $\mathbb{E}[\kappa(A_m)] = \Theta(m^{-1/2})$, $\mathbb{E}[\kappa(A_m)] = O(m^{-1/2})$ and $\mathbb{E}[\kappa(A_m)] \sim c m^{-1/2}$, so all of these are refuted too.

The construction of the norm matrix (the variety, the sampling law, the basis) is never used: the theorems below hold for *every* random matrix on *every* probability space, with the matrix size allowed to depend on m . Hence they apply to whatever norm matrix the conjecture intends. The remark about the logarithmic-potential integral describes how the expectation would be computed and does not change its value, so it is not used.

Conventions. Condition numbers take values in $[0, +\infty]$, with the standard convention $\kappa(A) = +\infty$ when A is singular (rank deficient). The expectation of a $[0, +\infty]$ -valued random variable is its Lebesgue integral. For measurable random variables this is the ordinary expectation; for arbitrary $[0, +\infty]$ -valued functions we use the lower Lebesgue integral (Mathlib’s `extttlintegral`), which always exists, so no integrability or measurability assumption is needed for the bound. A real-valued variant is in Theorem 7. Matrices are real and have at least one column ($n \geq 1$, resp. $q \geq 1$): an empty 0×0 matrix has $\|I\| = 0$ and is excluded. We use $m^{-1/2} = 1/\sqrt{m}$ for $m \geq 1$.

2 Condition numbers are at least 1

Definition 1 (norm condition number). Let R be a nontrivial normed ring, i.e. a ring with a norm satisfying $\|xy\| \leq \|x\| \|y\|$ and $1 \neq 0$. For $A \in R$ put $\kappa(A) = \|A\| \|A^{-1}\|$ if A is invertible and $\kappa(A) = +\infty$ otherwise. For real $n \times n$ matrices we use the spectral norm $\|A\|_2 = \sup_{\|x\|=1} \|Ax\|$ (Euclidean norms), which is submultiplicative; then $\kappa(A) = \|A\|_2 \|A^{-1}\|_2 = \sigma_{\max}(A)/\sigma_{\min}(A)$ for invertible A .

Definition 2 (singular-value condition number). For a real $p \times q$ matrix A put $\sigma_{\max}(A) = \sup_{\|x\|=1} \|Ax\|$ and $\sigma_{\min}(A) = \inf_{\|x\|=1} \|Ax\|$ (Euclidean norms, $x \in \mathbb{R}^q$). Define $\kappa_2(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$ if $\sigma_{\min}(A) > 0$ and $\kappa_2(A) = +\infty$ if $\sigma_{\min}(A) = 0$, i.e. if A does not have full

column rank. This covers rectangular norm matrices, for example an $m \times N$ matrix of evaluations at the samples.

Lemma 3. (a) In a nontrivial normed ring, $\kappa(A) \geq 1$ for every A . (b) For every real $p \times q$ matrix with $q \geq 1$, $\kappa_2(A) \geq 1$.

Proof. (a) If A is singular, $\kappa(A) = +\infty$. Otherwise $\|1\| = \|1 \cdot 1\| \leq \|1\|^2$ and $\|1\| > 0$ (as $1 \neq 0$) give $\|1\| \geq 1$, so $1 \leq \|1\| = \|AA^{-1}\| \leq \|A\| \|A^{-1}\|$. (b) If $\sigma_{\min}(A) = 0$, $\kappa_2(A) = +\infty$. Otherwise pick the unit vector $e_1 \in \mathbb{R}^q$: $\sigma_{\min}(A) \leq \|Ae_1\| \leq \sigma_{\max}(A)$, and $\sigma_{\min}(A)$ is finite and positive, so $\kappa_2(A) \geq \sigma_{\min}(A)/\sigma_{\max}(A) = 1$. $\square$

3 The refutation

Lemma 4. If (Ω, P) is a probability space and $X : \Omega \rightarrow [0, +\infty]$ satisfies $X \geq 1$ pointwise, then $\mathbb{E}[X] = \int X dP \geq \int 1 dP = 1$.

Theorem 5 (exact reading (E)). Let $m \geq 2$, let (Ω, P) be any probability space, and let $A : \Omega \rightarrow \mathbb{R}^{n \times n}$ ($n \geq 1$) or $A : \Omega \rightarrow \mathbb{R}^{p \times q}$ ($q \geq 1$) be any random matrix. Then $\mathbb{E}[\kappa(A)] \geq 1 > m^{-1/2}$, resp. $\mathbb{E}[\kappa_2(A)] \geq 1 > m^{-1/2}$. In particular neither expectation equals $m^{-1/2}$.

Proof. Lemmas 3 and 4 give the lower bound 1, and $m^{-1/2} < 1$ because $m > 1$ and the exponent is negative. $\square$

Theorem 6 (order reading (O)). Let (Ω_m, P_m) be probability spaces and A_m random $N_m \times N_m$ matrices ($N_m \geq 1$), resp. random $p_m \times q_m$ matrices ($q_m \geq 1$), indexed by the sample count m . There are no $C \in \mathbb{R}$ and M with $\mathbb{E}[\kappa(A_m)] \leq Cm^{-1/2}$ (resp. with κ_2) for all $m \geq M$.

Proof. Take $m = \max(M, \lceil C^2 \rceil + 1)$, so $m > C^2$. If $C \leq 0$ then $Cm^{-1/2} \leq 0 < 1$. If $C > 0$ then $C < \sqrt{m}$, so $Cm^{-1/2} = C/\sqrt{m} < 1$. In both cases $\mathbb{E} \geq 1 > Cm^{-1/2}$. $\square$

Theorem 7 (real-valued expectation). In any nontrivial normed ring, if A is almost surely invertible and $\|A\| \|A^{-1}\|$ is integrable, then $\int \|A\| \|A^{-1}\| dP \geq 1$.

Remark 8 (what is not refuted). At $m = 1$ we have $m^{-1/2} = 1$, which is attainable (e.g. $A = I$), so (E) is refuted for every $m \geq 2$ only. We do not address non-literal rewordings that change the quantity, such as “ $\mathbb{E}[\kappa] - 1$ is of order $m^{-1/2}$ ” (a deviation-from-isometry statement) or “ $\mathbb{E}[\log \kappa] = m^{-1/2}$ ”. The conjecture states the value of the expected condition number itself.

4 Relation to the earlier submission

PR #241 (orionsheep) gave the same argument ($\kappa \geq 1$, hence $\mathbb{E}[\kappa] \geq 1 > m^{-1/2}$), but it was closed without merging because its Lean capstone was the arithmetic fact $(1:\text{Nat}) < 2 \wedge 2 \neq 1$. The reviewer wrote that “the condition-number bound $\kappa \geq 1$ that constitutes the entire refutation of $\mathbb{E}[\kappa] = m^{-1/2}$ appears only in prose, with no matrices or expectations in Lean”. That PR also refuted only the exact value. The present Lean file defines both condition numbers on actual real matrices and proves $\kappa \geq 1$ from submultiplicativity and $\sigma_{\min} \leq \sigma_{\max}$. It formalizes the expectation as a Lebesgue integral over an arbitrary probability space and refutes both the exact reading at every $m \geq 2$ and the order reading for arbitrary m -indexed families.

5 Lean formalization

The file `lean/Conjecture2348/Basic.lean` (226 lines, Lean 4 / Mathlib v4.33.1, only import Mathlib) contains, in namespace C2348:

- `condNum` (norm condition number in a normed ring, $+\infty$ for non-units) and `one_le_condNum`; `condNum_matrix_eq` unfolds it, under the spectral norm (`Matrix.Norms.L2Operator`), to $\|A\|_2 \|A^{-1}\|_2$ with A^{-1} the matrix inverse if $\det A \neq 0$, and to $+\infty$ if $\det A = 0$;
- `stretchMax`, `stretchMin`, `svCond` ($\sigma_{\max}/\sigma_{\min}$ via `Matrix.toEuclideanLin`) and `one_le_svCond`;
- `one_le_expectation` (Lemma 4, with `lintegral`), `rpow_neg_half_lt_one`, `ne_rpow_neg_half`, `not_le_C_rpow_neg_half`;
- main theorems `conjecture_2348_false_exact`, `conjecture_2348_false_exact_sv` (Theorem 5), `conjecture_2348_false_order`, `conjecture_2348_false_order_sv` (Theorem 6) and `one_le_integral_condNum` (Theorem 7).

In the main theorems $m^{-1/2}$ is $(m:\mathbb{R}) \wedge (-(1/2:\mathbb{R}))$, embedded into $[0, +\infty]$ by `ENNReal.ofReal`. `lake build` succeeds, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for every main theorem. There is no `sorry` and no `native_decide`.